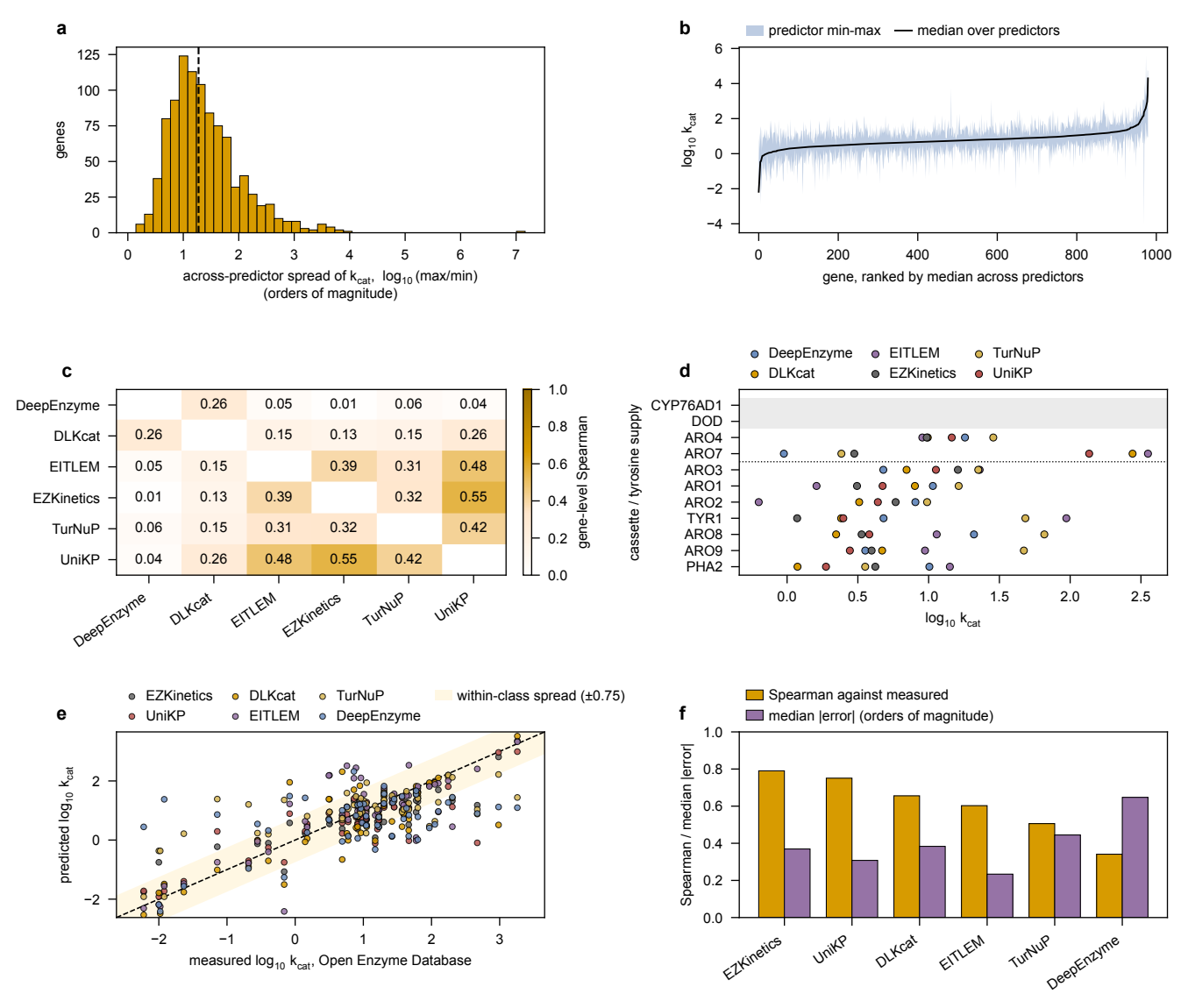

a
genes
across-predictor spread of k_{cat}, $\log_{10}$ (max/min)
(orders of magnitude)
b
predictor min-max
median over predictors
$\log_{10}$ k_{cat}
gene, ranked by median across predictors
c
DeepEnzyme
DLKcat
EITLEM
EZKinetics
TurNuP
UniKP
gene-level Spearman
d
DeepEnzyme
DLKcat
EITLEM
EZKinetics
TurNuP
UniKP
cassette / tyrosine supply
CYP76AD1
DOD
ARO4
ARO7
ARO3
ARO1
ARO2
TYR1
ARO8
ARO9
PHA2
$\log_{10}$ k_{cat}
e
EZKinetics
UniKP
DLKcat
EITLEM
TurNuP
DeepEnzyme
within-class spread (±0.75)
predicted $\log_{10}$ k_{cat}
measured $\log_{10}$ k_{cat}, Open Enzyme Database
f
Spearman against measured
median |error| (orders of magnitude)
Spearman / median |error|
EZKinetics
UniKP
DLKcat
EITLEM
TurNuP
DeepEnzyme